

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10

Year Plan

Class: 10

Level: Level 3

Academic Year: June 2026 - March 2027

Teaching Period: June - January (8 months)

Revision Period: February - March

ICT Teaching Resources Portal

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Academic Calendar Overview

The academic year for ICT Class 10 runs from June to March. The teaching period (June to January) covers the complete syllabus across 6 chapters at mastery level. Assessments are spread throughout the year: FA-1 in July, FA-2 in August, SA-1 (mid-year) in September, FA-3 in December, FA-4 in February, and SA-2 (annual board exam) in March.

Teaching Period

6 chapters are taught from June to December, with January reserved for revision and board exam preparation. Each month has approximately 12 periods. Emphasis is placed on in-depth understanding, problem-solving, and application-oriented learning.

Revision Period

January is dedicated to comprehensive revision and board exam preparation. February and March focus on intensive practice with previous board papers, model tests, and final consolidation with FA-4 in February and SA-2 (annual board exam) in March.

Monthly Teaching Plan

June	Ch 1: Advanced Programming Functions, File handling, Exception handling, Object-oriented concepts, Modular programming, Project work	12 periods 3 lab activities
July	Ch 2: Data Structures Arrays, Lists, Stacks, Queues, Dictionaries, Trees, Graphs, Sorting and searching algorithms, Time complexity	12 periods 3 lab activities
August	Ch 3: Database Fundamentals Database concepts, Normalization, Tables and relationships, Queries, Forms and reports, SQL basics, Advanced queries	12 periods 2 lab activities
September	Ch 4: Web Technologies HTML5, CSS3, JavaScript basics, Responsive design, Web page structure, Hosting and publishing, Web safety and ethics	12 periods 2 lab activities
October	Ch 5: Algorithms Algorithm design, Flowcharts, Pseudocode, Divide and conquer, Greedy algorithms, Dynamic programming, Computational thinking, Big-O notation	12 periods 3 lab activities
November	Ch 6: Emerging Technologies	12 periods 2 lab activities

Artificial intelligence, Machine learning basics, IoT concepts, Cloud computing, Blockchain fundamentals, Ethics in technology, Digital citizenship

December

Board Exam Revision - Chapters 1 to 3

12 periods

Concept review, Previous board question analysis, Practice worksheets, Lab re-runs, Doubt clarification, Problem-solving sessions

3 lab activities

January

Board Exam Preparation - Chapters 4 to 6 and Overall

12 periods

Previous board papers practice, Model tests, Lab viva preparation, Final revision, Exam strategy and time management

3 lab activities

Assessment Schedule

The assessment structure follows the continuous and comprehensive evaluation (CCE) pattern with Formative Assessments (FA), Summative Assessments (SA), Unit Tests, and Laboratory Examinations.

Assessment	Month	Marks	Topics Covered
FA-1	July	30	Ch 1: Advanced Programming
Unit Test 1	July	10	Ch 1: Advanced Programming
FA-2	August	30	Ch 2 & 3: Data Structures, Database Fundamentals
Unit Test 2	August	10	Ch 2 & 3: Data Structures, Database Fundamentals
SA-1	September	40	Ch 1 to 3: Half Yearly
Lab Exam 1	September	20	Practical: Ch 1 to 3
FA-3	December	30	Ch 4 & 5: Web Technologies, Algorithms
Unit Test 3	December	10	Ch 4 & 5: Web Technologies, Algorithms
FA-4	February	30	Ch 5 & 6: Algorithms, Emerging Technologies
Unit Test 4	February	10	Ch 5 & 6: Algorithms, Emerging Technologies
SA-2	March	40	Ch 1 to 6: Annual Board Examination

Lab Exam 2	March	20	Practical: Ch 1 to 6
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Resources and Materials

The following resources are required for effective delivery of the ICT curriculum for Class 10 at the mastery level:

- **Textbooks:** NCERT ICT textbook for Class 10, KREIS supplementary ICT workbook, Board exam preparation guide
- **Software:** Python 3.x, Scratch 3.0, LibreOffice Base, LibreOffice Writer, LibreOffice Calc, VS Code, Web browsers (Chrome/Firefox), Database management tools, Online IDEs (Replit, Trinket)
- **Hardware:** Desktop computers or laptops (minimum 1 per student), projector and screen, speakers and headphones, internet connectivity, USB storage devices
- **Laboratory:** ICT laboratory with adequate power supply, UPS backup, networked computers with domain login, wall charts displaying Python syntax, algorithm symbols, HTML tags, and database schemas
- **Reference Materials:** KREIS ICT teacher handbook, previous year board examination papers (2018-2025), solved board exam papers, online coding platforms, Python documentation, W3Schools web development tutorials, KREIS practice question banks

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Note: This year plan is designed for **Level 3 Mastery / Board Exam Prep** focus. The curriculum emphasizes in-depth understanding of programming concepts, advanced data structures and database management, comprehensive web development skills, and algorithmic thinking, preparing students for board examinations and higher secondary ICT studies.